Could It Be Chance?

A plain-language guide to hypothesis testing

Data Analysis in R using Agentic AI Assistants, Session 4

With one example throughout: the Lab 2 claim about conflict and income

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Key concept: red boxes

The idea in plain words. Read only the red boxes and you have the argument.

In technical terms: grey boxes

The same step with a formula and the line of R that computes it. Optional on first reading.

Try it: green boxes

Something to run yourself, in the Console, without the assistant.

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How to use this guide

In Session 2 you were given a claim:

Since the end of the Cold War, internal armed conflict has been concentrated in poorer countries.

You built a table with one row per country and year, and in Session 3 you turned it into figures. Every figure that came back shows the same thing: poorer countries are in conflict much more often than rich ones. This guide asks the next question. **Could a pattern like that be produced by chance?** And if not by chance, how sure can we be about its size?

Those are the questions a **hypothesis test** answers. The guide assumes no mathematics. Everything is explained in words, with pictures, and with numbers from the table you already have, `lab2_full_table.csv`. Every number in it was computed from that file. The script that draws the figures is `figures_ht.R`; the script we use in class is `scripts/session4_tests.R`. Both are in the course folder.

The guide uses three kinds of boxes.

Key concept: what a red box is

A red box states one idea you need to carry forward, in plain words. The red boxes are the backbone of the guide.

In technical terms: what a grey box is

A grey box says the same thing in the language of the readings (Kennedy and Waggoner 2021, ch. 7, sections 7.1–7.3; Imai 2017, ch. 7; Dean and Illowsky 2018, chs. 9–10), usually with an equation and the line of R that computes the number. You do not need to manipulate the equations. You need to *recognise* them, so that the output of `t.test()` and the text your assistant writes stop looking like a foreign language. A decoder at the end lists the symbols.

Try it: what a green box is

A green box is something to type in the Console. Do these yourself, not through the assistant. The point is to see the numbers move.

1 Where we are

1.1 Concentrated compared with what?

Start with the most natural way of checking the claim: take all the country-years with an internal conflict and ask where they are. Of the 1,004 conflict country-years between 1991 and 2025 for which we know the previous year's income, 68% are in countries with an income below $10,000 per head. Half (51%) are below $5,000. That sounds like strong support: conflict is concentrated in poor countries.

It is not enough. Most country-years *of any kind* are in poorer countries, because most countries are not rich. Of the 4,775 country-years *without* conflict, 39% are also below $10,000 and 24% below $5,000. Peaceful years are concentrated in poorer countries too. So the question is not whether conflict years pile up among the poor, but whether they pile up *more* than peaceful years do. Figure 1 puts the two side by side.

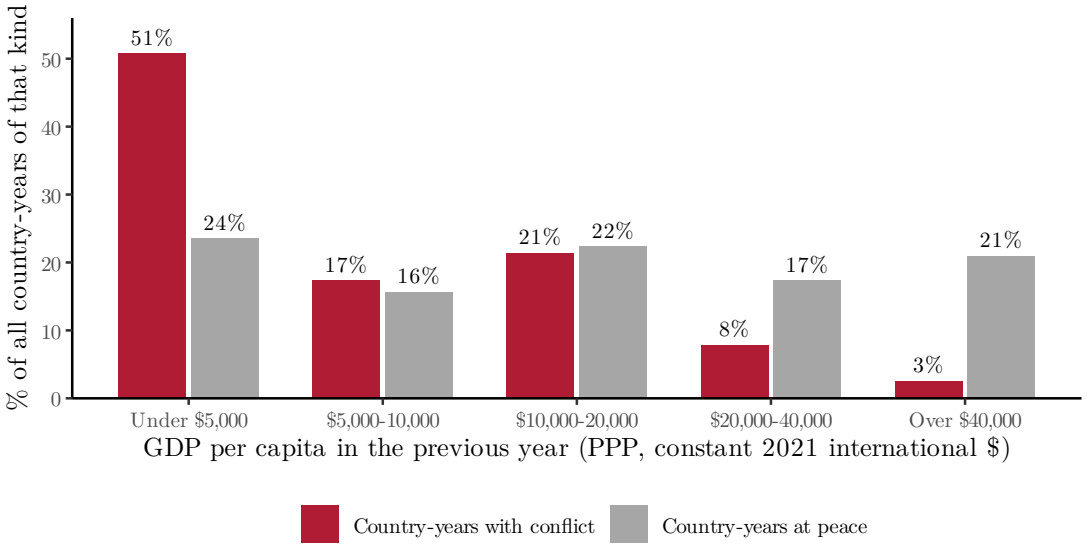


Figure 1: Where conflict years and peaceful years are, by income in the previous year, 1991–2025. Each colour adds up to 100%. Red: the 1,004 country-years with an internal conflict. Grey: the 4,775 without.

Now the claim has something to stand on. Below $5,000, the red bar is twice the grey one (51% against 24%); above $40,000, it is an eighth of it (3% against 21%). Between $5,000 and $20,000 the two bars are about the same height: conflict years are no more concentrated there than peaceful years are.

The same comparison can be read the other way round. Instead of asking where the conflict years are, ask how often country-years in each income band are in conflict. Below $5,000: 31.2%. Over $40,000: 2.5%. In between: 18.9%, 16.7% and 8.7%. The two readings are the same information. "Conflict years are more concentrated among the poor than peaceful years are" and "poor country-years are in conflict more often" are true or false together.

Key concept: concentration is a comparison

"Most conflicts happen in poor countries" proves nothing on its own, because most of everything happens in poor countries: there are more of them. A claim about concentration is a claim about a *difference*: conflict years must be concentrated among the poor *more than* peaceful years are. Every test of the claim needs the peaceful years in it. That is

why a table of conflict country-years alone (Session 2) could not test the claim, and why a histogram of the incomes of conflict years alone (Figure 1 of the Session 3 gallery) shows nothing.

1.2 One version of the test

We did not settle on one test, or on one figure, in class. The five figures that came back made different choices, and Section 8 shows how much those choices matter. To have numbers we can follow from section to section, this guide uses one version throughout. It is a defensible version, not the only one:

- **One row** is a country in a year, 1991–2025. Peaceful years are in the table, coded 0.
- **Conflict** means that the UCDP/PRIO Armed Conflict Dataset (Gleditsch et al. 2002; version 26.1) records an internal armed conflict (type 3 or 4) located in that country in that year: at least 25 battle-related deaths in the year.
- **Poorer** means income in the *previous* year, because war lowers income and we do not want the war to explain itself.
- Within each year, countries are split into **four equal groups** by that income, from the poorest quarter to the richest. Within each year, because almost every country got richer between 1991 and 2025.

276 country-years have no income for the previous year and are dropped; 5,779 remain, from 167 countries.

Four equal groups make the comparison of Figure 1 automatic. Each quarter holds a quarter of all country-years. If income had nothing to do with conflict, each would also hold about a quarter of the conflict years. In fact the poorest quarter holds 48% of the conflict years and 20% of the peaceful ones; the richest holds 7% of the conflict years and 29% of the peaceful ones. Figure 2 shows the same thing as shares in conflict.

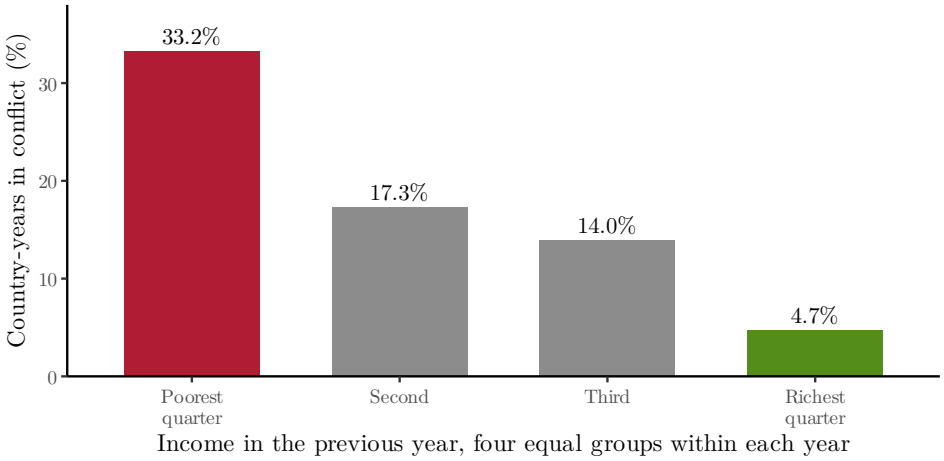


Figure 2: Share of country-years with an internal armed conflict, by income in the previous year, 1991–2025. 5,779 country-years, 167 countries.

In the poorest quarter, 33.2% of country-years see an internal conflict; in the richest quarter, 4.7%. The **gap** is 28.5 percentage points. That gap is the number we will test throughout this guide.

1.3 What your figures agree on

The five figures that came back after Session 3 made different choices, and none of them is *the* answer. Some used today's World Bank income groups, some built quarters of their own; some used income in the same year, one filled in the missing incomes; two counted the *number* of conflicts per country rather than whether a country was in conflict. Yet every figure shows poorer countries on top and the richest at the bottom, and every one of them (unlike the tables from Session 2) compares conflict years with peaceful years, either by plotting shares or by averaging over all country-years. Section 8 comes back to how much those choices matter. They change the size of the gap, but not its direction.

So the pattern is not in doubt. What a figure cannot tell you is whether a pattern of this size could have come about by chance, and how far the true gap could be from 28.5. A figure also invites you to see things that may not be there. Two of your figures said that the gap between poor and rich countries is *widening*. Is it? Section 7 tests that too.

Key concept: what a figure cannot do

A figure shows a pattern. It cannot tell you whether chance alone could have produced a pattern like it. Every pattern in data is partly signal and partly noise, and the eye is very bad at telling which is which. A hypothesis test is a way of asking, with numbers: how big is the noise?

2 What could "chance" mean here?

2.1 We have all the countries

In a survey, chance has an obvious meaning. We interview 1,000 people out of millions; a different 1,000 would have given a slightly different answer. That is **sampling variation**, and standard errors measure it.

Our table is different. It contains (almost) every country in the world in every year. There is no other sample we could have drawn. So what could chance possibly mean?

The usual answer is a thought experiment. Think of the conflicts we observe as one run of a process that could have gone differently: a coup that failed by an hour, a rebel leader who died in a car crash, a drought that came a year later, an election that went the other way. Rerun the 1990s and 2000s a thousand times and the world would not look exactly the same each time. Some country-years that were at peace would see a conflict, and the other way round. The question then is whether the gap between poor and rich countries is a stable feature of the process — something that would show up in most reruns — or something that just happened this time.

Key concept: chance without a sample

Even with data on every country, the world we observe is one of many ways history could have gone. A hypothesis test asks: if income had nothing to do with conflict, how often would a world like ours produce a gap as big as the one we see? That is a question about the *process* that produces conflict, not about which countries we happened to sample.

You do not need to settle this philosophical question to use the tools, but you should know that the tools rest on *some* story about where chance comes from. Section 6 shows that the story matters a great deal for the answer.

3 The logic of a test

Every hypothesis test has the same five steps.

1. **Say what "nothing going on" means.** Here: income has nothing to do with conflict. In the language of Section 1: conflict years are spread across income levels *just like* peaceful years, so each quarter holds about a quarter of them. This is the **null hypothesis**.
2. **Work out what the world would look like if it were true.** In particular: how big a gap would chance alone produce?
3. **Compare.** Put the gap we actually see next to the gaps chance produces.
4. **The p-value.** If nothing were going on, how often would we see a gap at least as big as ours?
5. **Decide.** If that is rare enough — by convention, less than 5% of the time — we *reject* "nothing going on".

Notice what the test can and cannot do. It can reject the null. It can never prove it. A test that does not reject says "not enough evidence", never "no effect".

The whole difficulty is in step 2. The next two sections show two ways of doing it: by shuffling, and by formula.

4 A test you could do with a deck of cards

4.1 Shuffling the labels

If income had nothing to do with conflict, then the labels "poorest quarter", "second", "third" and "richest quarter" would be arbitrary. Any country-year could just as well carry any label. So let us *make* them arbitrary. In each year, take the 160 or so income labels, shuffle them like a deck of cards, and deal them out again at random. Then compute the share in conflict in each group, and the gap between the poorest and the richest.

Figure 3 shows the result of three such shuffles next to what we actually observe. In the shuffled worlds, every group is in conflict about 17% of the time, which is the average for all country-years. The groups differ by a point or two, in no particular direction. The real world looks nothing like that.

Three shuffles are not enough to know what chance can do. A computer can shuffle 2,000 times in a few seconds. Figure 4 shows the 2,000 gaps.

The shuffled gaps pile up around zero, as they should: when the labels mean nothing, the poorest group is no more likely to be in conflict than the richest. They spread out a little, typically by about 1.4 points either way, and the largest of the 2,000 is 4.9 points. Our gap is 28.5. Not one shuffle comes close.

4.2 The p-value

The **p-value** is the share of shuffles that produce a gap at least as big as the one we observe. Here it is 0 out of 2,000. We would write $p < 0.001$: if income had nothing to do with conflict, a gap like ours would come up less than once in a thousand reruns. We reject the null hypothesis.

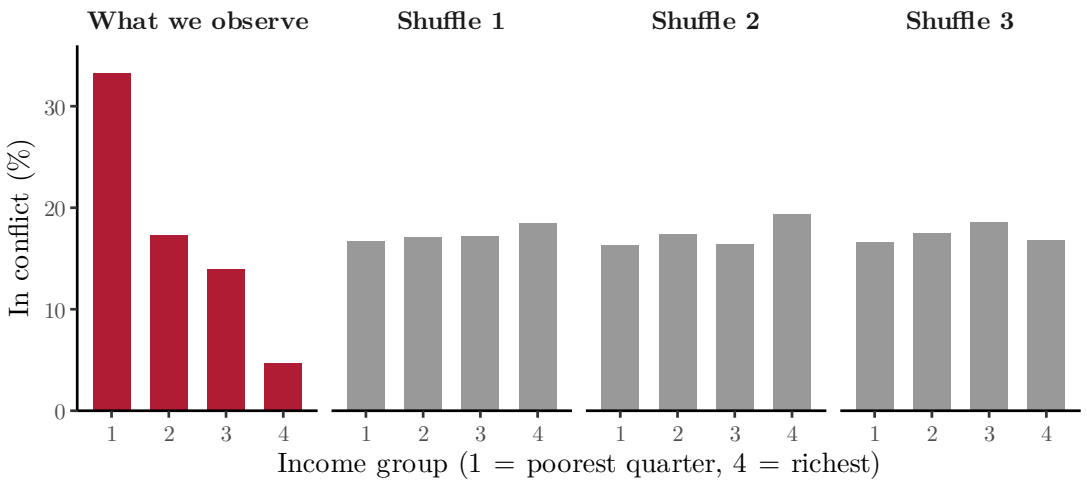


Figure 3: What we observe (left) and three worlds in which income has nothing to do with conflict, made by shuffling the income labels within each year.

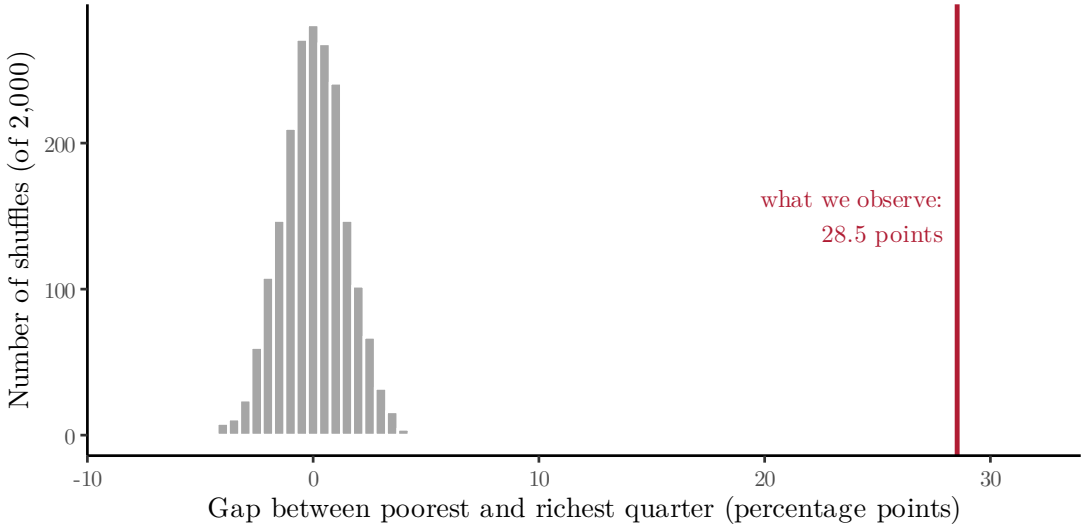


Figure 4: 2,000 gaps between the poorest and the richest quarter, each from one shuffle of the income labels. Red line: the gap we observe.

Key concept: the *p*-value

The *p*-value answers one question: *if nothing were going on, how often would we see a result at least this extreme?* A small *p*-value says that our result would be rare in a world where nothing is going on. That is all it says.

The American Statistical Association felt the need to issue a statement on what *p*-values do and do not mean (Wasserstein and Lazar 2016). The three most common misreadings:

- It is **not** the probability that the null hypothesis is true. It is calculated *assuming* the null is true.
- It is **not** the size or the importance of the gap. A tiny gap in a huge dataset can have a tiny *p*-value.
- It is **not** a measure of whether the gap is *caused* by income. Section 9 returns to this.

In technical terms: permutation tests

What we just did is a **permutation test** (also called a randomisation test). The collection of statistics computed from the shuffled data is the **null distribution** or **permutation distribution**. With R shuffles and b of them at least as extreme as the observed statistic, the p-value is usually reported as $(b+1)/(R+1)$, so with $b = 0$ and $R = 2{,}000$ we report $p < 0.0005$. The test is two-sided if we count shuffled gaps as extreme in either direction, `abs(gap_shuffled) >= abs(gap)`. In R:

```
for (i in 1:1000) {
  d_shuffled <- d_test %>% group_by(year) %>%
    mutate(group_shuffled = sample(income_group)) %>% ungroup()
  poor <- mean(d_shuffled$conflict[d_shuffled$group_shuffled == 1])
  rich <- mean(d_shuffled$conflict[d_shuffled$group_shuffled == 4])
  shuffled_gaps[i] <- poor - rich
}
mean(abs(shuffled_gaps) >= gap)     # the p-value
```

`sample(x)` returns the values of `x` in random order. Shuffling *within* each year keeps the number of countries in each group in each year the same as in the real data.

Try it: shuffle once by hand

Run section 2 of `session4_tests.R`, but first replace `1000` by `1` and print `d_shuffled %>% count(income_group, group_shuffled)`. What has the shuffle done to the labels? Then run the loop with 1,000 and look at `summary(shuffled_gaps)`.

5 The same test from a formula

5.1 The standard error

Shuffling is the most transparent way of finding out what chance can do, but statisticians found a short cut long before computers. The spread of the shuffled gaps in Figure 4 can be predicted with a formula. That spread is called the **standard error** of the gap: the typical distance between the gap we compute and the gap a different run of history would give.

For a share, the formula says in words: *the standard error is larger when the share is near 50% and smaller when there are more observations.* For the gap between two shares we combine the two. With 1,460 country-years in the poorest quarter (33.2% in conflict) and 1,428 in the richest (4.7%), the formula gives a standard error of 1.35 points. The shuffles gave 1.40. Formula and shuffle agree, up to the small wobble of any simulation.

5.2 How many standard errors away?

Now the test becomes a single division. Our gap is 28.5 points; chance moves it by about 1.35 points. So the gap is $28.5/1.35 = 21$ standard errors away from zero. Under the bell curve, anything beyond about 2 standard errors happens less than 5% of the time. Twenty-one is off the chart.

Key concept: the test statistic

Every test in this course asks the same question: *How many standard errors away from "nothing" is my estimate?* That number is the **test statistic** (z or t). A gap of 28 points means nothing on its own. If chance moves gaps by 1 point, it is enormous; if chance moved them by 20 points, it would be nothing special. Beyond about 2 standard errors, p

is below 0.05.

5.3 An interval, not just a verdict

Turn the same calculation around and you get a **95% confidence interval**: the gap plus or minus about two standard errors. Here: from 25.9 to 31.2 points. Intervals built this way capture the true gap in 95% of reruns. The interval tells you more than the p-value. It says not only "not zero" but also "somewhere between 26 and 31 points" — if the formula is right. Section 6 explains why, here, it is not.

In technical terms: two-sample test for proportions

For two groups with shares $\hat{p}_1$ and $\hat{p}_2$ based on n_1 and n_2 observations,

$$\mathrm{SE}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} = \sqrt{\frac{0.332 \times 0.668}{1460} + \frac{0.047 \times 0.953}{1428}} = 0.0135,$$

the test statistic is $z = (\hat{p}_1 - \hat{p}_2)/\mathrm{SE} = 0.285/0.0135 = 21.1$, and the 95% interval is $(\hat{p}_1 - \hat{p}_2) \pm 1.96 \times \mathrm{SE}$. R's `prop.test()` does the same calculation (with a small "continuity correction") and reports $\chi^2 = z^2$:

```
prop.test(c(485, 67), c(1460, 1428))
# X-squared = 378.16, df = 1, p-value < 2.2e-16
# 95 percent confidence interval: 0.258 0.312
```

`p-value < 2.2e-16` is R's way of saying "smaller than the smallest number I bother to print". Report it as $p < 0.001$.

5.4 All four groups at once

The gap compares only the poorest and the richest quarter. The **chi-square test** for a contingency table (Kennedy and Waggoner 2021, section 7.3) asks whether conflict is spread evenly across *all four* groups. It compares the counts we see with the counts we would expect if conflict had nothing to do with income. If income did not matter, each quarter would have about 250 conflict country-years (17.4% of its rows). We see 485, 251, 201 and 67.

In technical terms: the chi-square test

For a table with observed counts O and the counts E expected under independence,

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad df = (\text{rows} - 1)(\text{columns} - 1).$$

In R: `chisq.test(table(d_test$income_group, d_test$conflict))` gives $\chi^2 = 427.1$ with $df = 3$, $p < 0.001$. `chisq.test(...)$expected` prints the expected counts. The chi-square test says only that the groups differ *somehow*; it does not say which way. Always look at the shares as well.

Try it: formula versus shuffle

Run section 3 of the script. Compare `se` with `sd(shuffled_gaps)` from section 2. Then compare the interval from `prop.test()` with `gap ± 1.96 * sd(shuffled_gaps)`.

6 The catch: country-years are not independent

6.1 What the shuffle assumed

Both the shuffle and the formula rest on one assumption, and it is easy to miss. The shuffle dealt out labels to *single country-years*, as if each year of each country were a separate draw from the process that produces conflict. The formula assumed the same: 1,460 independent observations in the poorest quarter.

Look at Figure 5. Each row is a country, each square a year, red means conflict. Conflict comes in long runs. Afghanistan, Colombia, India, Myanmar and the Philippines are in conflict in every one of the 35 years. Most countries are never in conflict at all: only 80 of 173 countries see any internal conflict between 1991 and 2025. A country in conflict this year is in conflict next year 85% of the time; a country at peace this year starts a conflict next year 3% of the time. And 20 countries supply more than half (55%) of all conflict country-years.

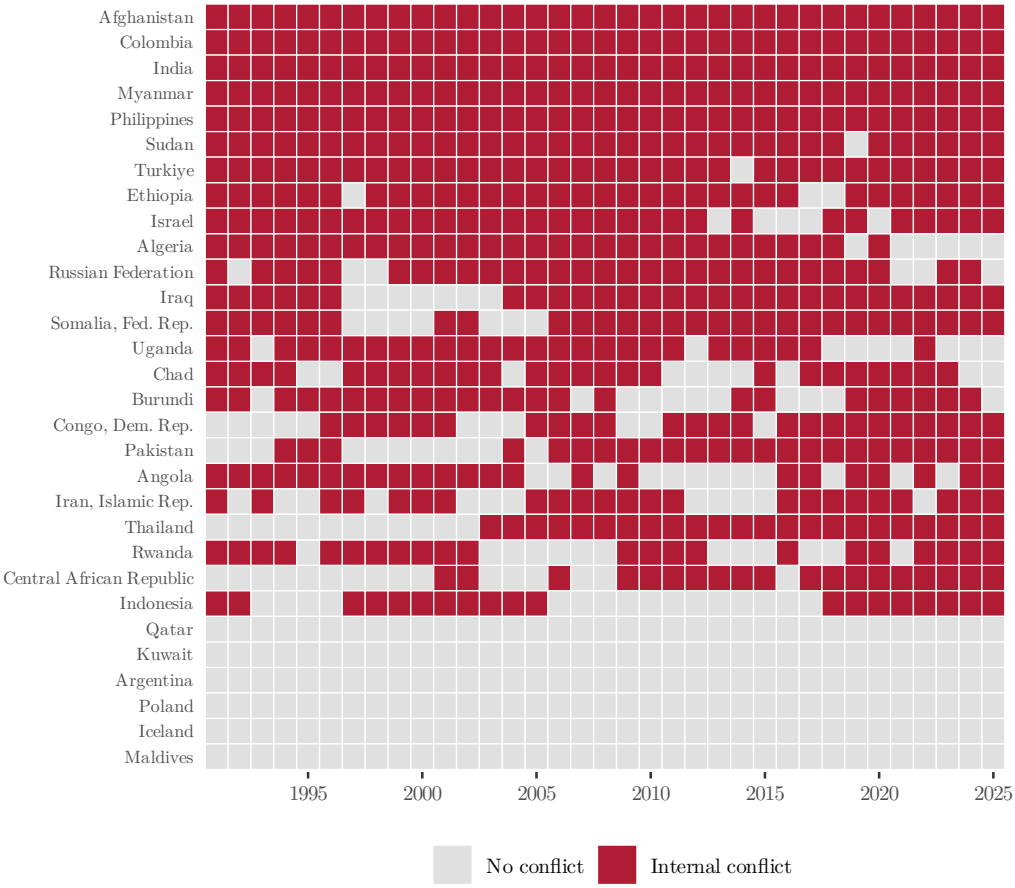


Figure 5: Conflict by year for the 24 countries with the most conflict years, 1991–2025, and six countries without any. Red: internal armed conflict.

So Afghanistan in 2014 and Afghanistan in 2015 are not two independent pieces of evidence. They are, in large part, the same piece of evidence counted twice. Our 5,779 country-years contain much less information than 5,779 independent observations would.

Key concept: count what is independent

A test is only as good as its idea of what varies by chance. If the same countries appear year after year and their conflicts last for years, the independent pieces of evidence are closer to *countries* than to country-years. Treating country-years as independent pretends we have far more evidence than we do, and makes every interval too narrow and every p-value too small.

6.2 Shuffle whole countries

The fix is to shuffle what is actually independent. Keep each country's conflict history as it is, and give it the *whole* income history of another country, chosen at random. A shuffled world might give Afghanistan's 35 years of conflict the income path of Norway, and Norway's 35 peaceful years the income path of Malawi. Then recompute the four groups, year by year, and the gap. Repeat 2,000 times.

Figure 6 compares the two kinds of shuffle. Shuffling whole countries produces much larger gaps by chance. The spread (the standard error) is 6.0 points instead of 1.4: more than four times as large. Chance alone now produces gaps of 12 points or more (in either direction) about one time in twenty, and the largest of the 2,000 is 21 points.

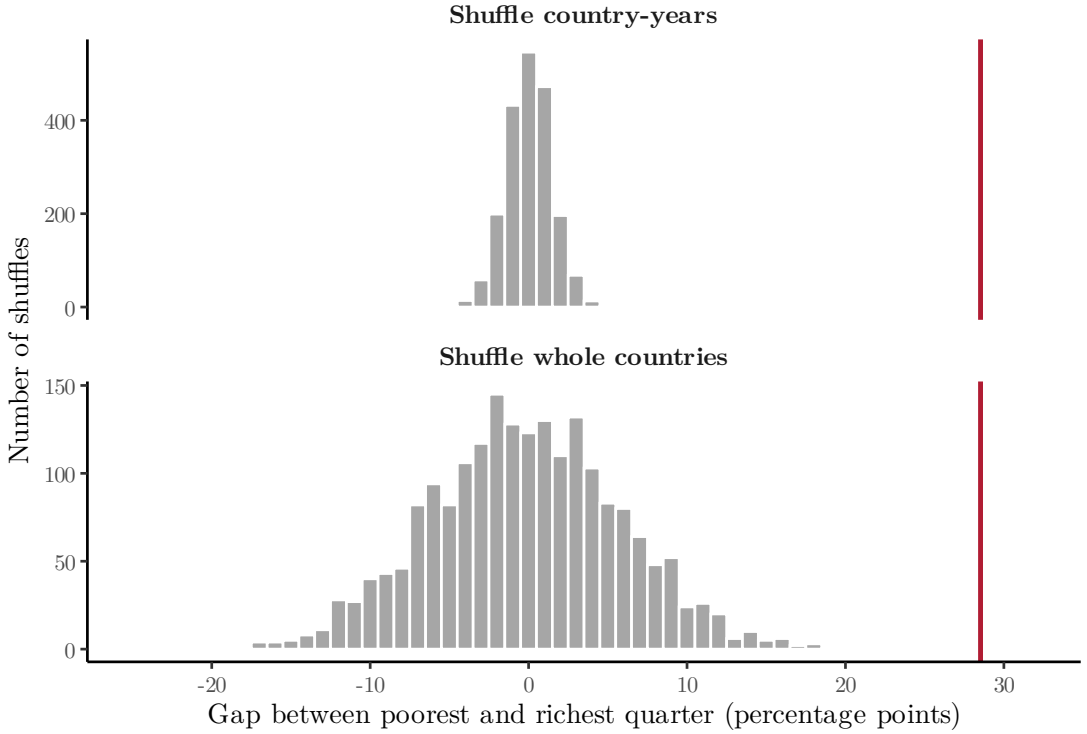


Figure 6: What chance can do, two ways. Top: shuffling single country-years, as in Figure 4. Bottom: shuffling whole countries, so that each country keeps its own conflict history. Red line: the gap we observe, 28.5 points.

And yet: our gap of 28.5 points is still beyond every one of the 2,000 country shuffles. The claim survives. What changes is how precisely we know the size of the gap.

6.3 The honest interval, three ways

Figure 7 puts the naive interval next to three honest ones.

- **Formula, country-years as independent:** 25.9 to 31.2 points. Too narrow.
- **Shuffle whole countries:** gap ± 1.96 × 6.0, from 16.7 to 40.4 points.
- **Clustered standard errors:** a formula that allows the years of the same country to be related (grey box below). From 17.9 to 39.1 points.
- **Countries as the unit:** collapse the table to one row per country (the share of its years in conflict, and its median income), split the 167 countries into four groups, and compare the poorest quarter of countries (in conflict 35.0% of the time on average) with the richest (5.6%) with a *t*-test (Kennedy and Waggoner 2021, section 7.2). The gap is 29.4 points, from 17.2 to 41.6.

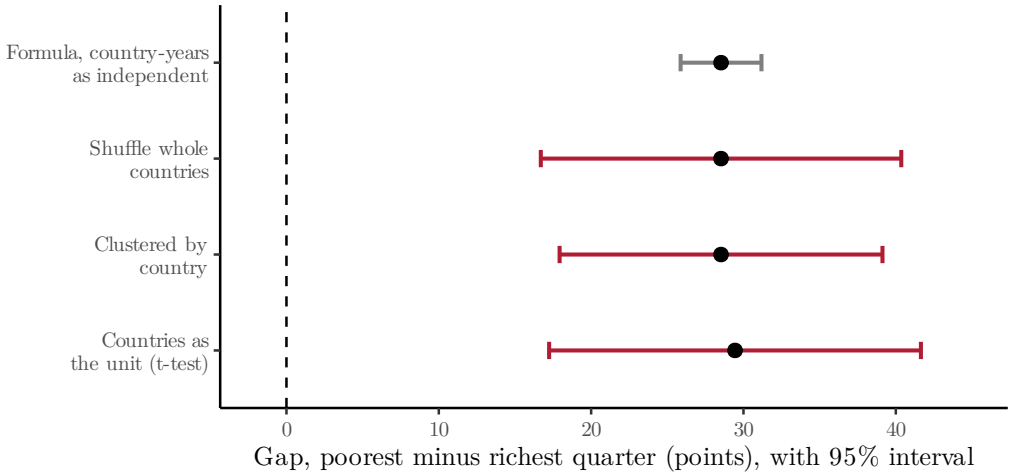


Figure 7: The gap between the poorest and the richest quarter, with 95% intervals computed four ways. Only the top one treats country-years as independent.

The three honest methods agree with each other, and all disagree with the naive formula. The honest interval is about four times as wide. Put differently: the 2,888 country-years in the two outer quarters carry about as much information as 180 independent observations would.

Key concept: a correct formula with the wrong assumption

The formula for the standard error was correct, and R computed it correctly. It was the wrong formula for these data, because it assumes independent observations. Nothing in the output warns you. `prop.test()`, a Wilson interval, a logistic regression: all report precise-looking numbers based on the same assumption. Ask of every standard error: *what does it assume varies by chance?*

In technical terms: clustered standard errors and the design effect

When observations come in groups (**clusters**: here, countries) and are related within groups, the usual standard errors are too small. **Cluster-robust** standard errors allow any pattern of dependence within a cluster and assume independence only between clusters. In R, with the `sandwich` and `lmtest` packages, regress the 0/1 outcome on a 0/1 indicator for the poorest group (a *linear probability model*: its slope is exactly the gap in shares, see Session 6):

```
m <- lm(conflict ~ poor, data = d_two_groups)          # poor = 1, richest = 0
coeftest(m)                                            # SE 1.36 points
coeftest(m, vcov = vcovCL(m, cluster = ~ iso3c))       # SE 5.41 points
```

The squared ratio of the two standard errors, $(5.41/1.36)^2 \approx 16$, is the **design effect**: the factor by which dependence inflates the variance. Dividing the number of observations by it gives the **effective sample size**, here about $2{,}888/16 \approx 180$. Clustered standard errors need a reasonable number of clusters; with 160 countries we have plenty.

Try it: shuffle whole countries

Run sections 4 and 5 of the script. Compare `sd(shuffled_gaps_countries)` with `sd(shuffled_gaps)`. Then run section 6. Does the country-level t-test agree with the country shuffle?

7 Testing what a figure suggested: is the gap widening?

Two of your figures carried a second claim: the gap between poor and rich countries has widened since the 1990s. In one, the share of low-income countries in conflict climbs from about 40% in the 1990s to 60% in the 2020s, while the high-income share falls. It looks dramatic.

Now treat it as a hypothesis. The null: the gap is the same in 1991–2005 as in 2011–2025. The test statistic: the change in the gap.

With our test (income in the previous year, quarters within each year), the gap is 27.8 points in 1991–2005 and 32.1 points in 2011–2025: a change of 4.3 points. Is that more than chance? Shuffle whole countries again, and each time compute the change. Figure 8 shows the answer. Changes of 4 points or more, in either direction, come up in about half of the shuffles ($p = 0.52$; with clustered standard errors, $p = 0.51$). There is no evidence that the gap has changed.

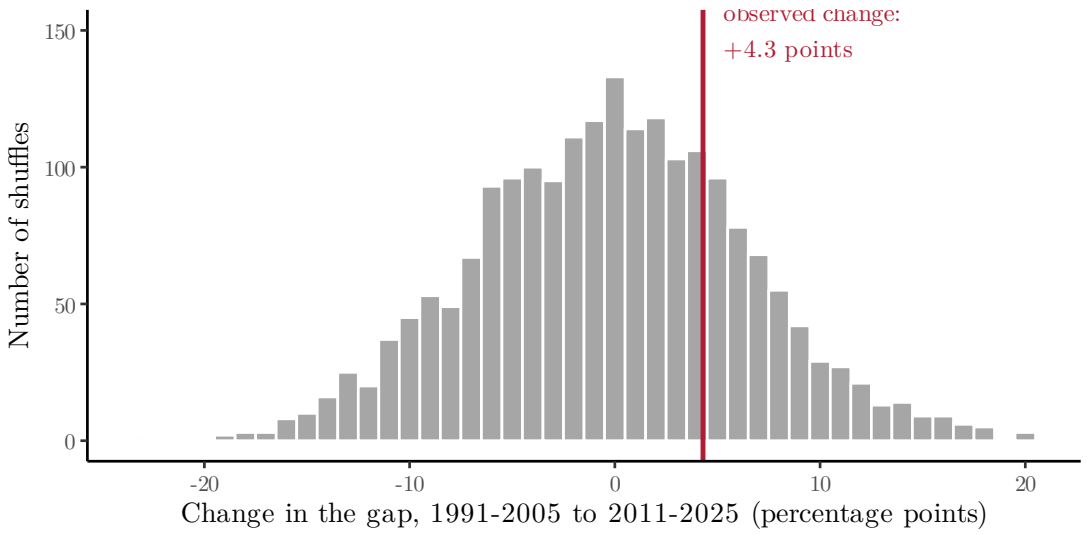


Figure 8: Change in the gap between the poorest and the richest quarter, from 1991–2005 to 2011–2025, in 2,000 country shuffles. Red line: the change we observe.

Why did the figures look so different? They used the World Bank's *current* income groups, applied to every year since 1990. With those groups, the low-income share rises from 37.9% to 54.1% and the change in the gap is 17.4 points. But look at which countries drive the rise: Syria, Yemen, South Sudan, the Central African Republic, Mali, Burkina Faso, Mozambique.

Most were not low-income countries in the 1990s. They are low-income *now*, in good part *because* of the wars that started after 2010. Classifying countries by today's income puts the effect of recent wars into the measure of poverty. Even so, with honest standard errors, the change is 17.4 points $\pm$ 2 $\times$ 9.5: $p = 0.07$, not beyond what chance could produce.

Key concept: a pattern you noticed is a hypothesis

A pattern you spot in a figure is a hypothesis, not a finding. Before you put it in a title, ask what the null would be, test it with standard errors that respect the structure of the data, and check that the measure does not build the result in. Be especially careful with patterns you noticed *after* looking: the eye picks out the most striking feature of whatever it is shown, and almost any dataset has some striking feature.

8 Many tests, many choices

8.1 One test per year

One of your scripts asked whether the poorest group comes out on top *in every single year*. In that figure it does in 33 of 36 years. Does that count against the claim?

Figure 9 runs the year-by-year version of our test: the gap between the poorest and the richest quarter, separately for each of the 35 years. In every year the gap is positive, between 16.5 and 47.6 points. In 33 years it is "significant" at 5%; in 2009 and 2012 it is not. That does not mean the relationship disappeared in 2009. Each yearly test rests on only about 42 countries per quarter, so its interval is wide, and the two grey points are simply the years in which chance pushed the gap down a little.

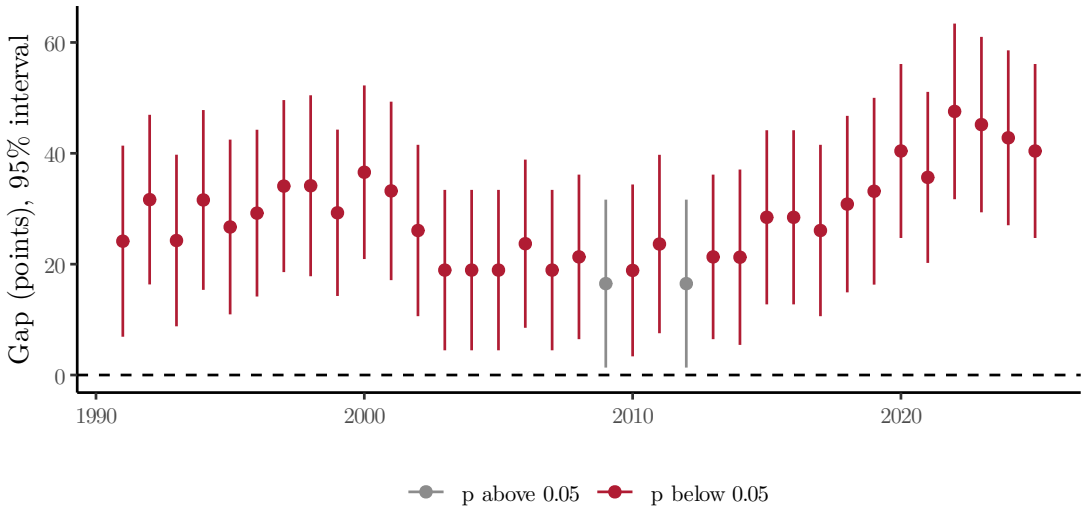


Figure 9: The gap between the poorest and the richest quarter, year by year, with 95% intervals. Red: $p < 0.05$ in that year's test.

The general lesson runs both ways. Run many tests where nothing is going on, and some will come out "significant" by chance: with 30 independent tests at 5%, the chance of at least one false alarm is $1 - 0.95^{30} = 79\%$. Run many small tests where something *is* going on, and some will miss it. A rule like "it must hold in every year" sounds strict but is mostly a rule about noise.

In technical terms: error rates and multiple testing

A **Type I error** (false alarm) is rejecting a true null; we choose its rate α, usually 0.05. A **Type II error** (missed effect) is failing to reject a false null; its rate depends on the size of the effect, the noise and the number of observations, and 1− its rate is the test's **power**. With k independent tests of true nulls, Pr(at least one false alarm) $= 1 - (1 - \alpha)^k$: 0.40 for $k = 10$, 0.79 for $k = 30$. The simplest correction (Bonferroni) tests each at α/k. The better habit is to decide in advance which *one* test answers your question.

8.2 The garden of forking paths

Before a single test was run, each of you made a dozen choices: which year's income, which groups, what to do with missing income, what counts as conflict. Figure 10 shows the gap, with honest (clustered) intervals, under the choices that came back in your scripts.

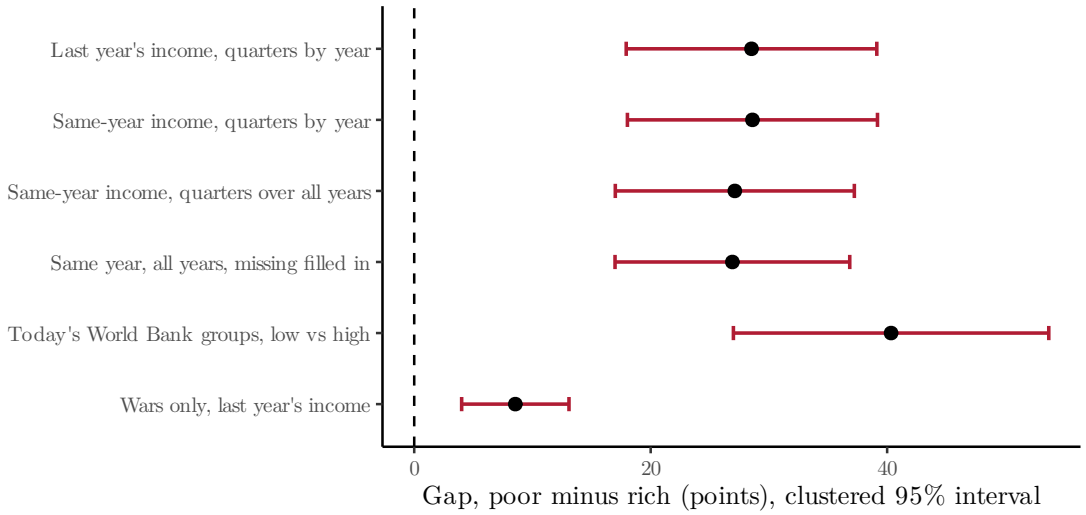


Figure 10: The gap between poor and rich country-years under six sets of choices, with clustered 95% intervals. The first row is the test of this guide.

Three things stand out.

- The reasonable choices about income (previous or same year, quarters within years or over all years, filling in missing income or not) barely move the gap: 27 to 29 points. That is good news for the claim. A finding that survives every reasonable choice is a robust finding.
- Today's World Bank groups give a much larger gap (40 points), for the reason explained in Section 7: they classify countries partly by the results of their wars.
- Counting only wars (1,000 or more battle deaths in the year) gives a much smaller gap, 8.6 points, because wars are rarer. It is still clearly above zero.

The danger is not in making choices; you have to. The danger is in making them *after* seeing which one gives the most striking result. Statisticians call this the **garden of forking paths** (Gelman and Loken 2014). With an AI assistant, it is easier than ever to walk down every path in a minute and report the prettiest. A p-value from the prettiest path means nothing, because the path was chosen by the data.

Key concept: decide first, then test

Write down your test before you run it: the unit, the outcome, the groups, the comparison, and what result would count against the claim. That is what Step 5 of Lab 2 asked you to do in `notes/lab2.md`. Then report the test you planned, and show the others as checks. If the answer changes with the path, say so: that is a finding too.

9 What a test cannot tell you

A hypothesis test answers one narrow question: could chance produce a result like this? It says nothing about several things that matter more.

Size and importance. With enough data, trivial differences become "significant". Always report the estimate and its interval first. "Poor country-years are in conflict about 29 points more often than rich ones, give or take 11" says more than "$p < 0.001$".

Cause. The test says the gap is not chance. It does not say that poverty *causes* conflict. Using last year's income helps against one problem — war lowering income — but conflicts last for years, so last year's income can already be shaped by the same conflict. Poor countries also differ from rich ones in a hundred other ways: weak states, rough terrain, large populations, colonial histories (Fearon and Laitin 2003; Collier and Hoeffler 2004). Separating income from those is a problem of research design, not of chance. One well-known attempt uses rainfall, which shifts the incomes of farming economies for reasons that have nothing to do with politics (Miguel, Satyanath and Sergenti 2004). Sessions 6 and 8 come back to it.

Bad measures. A test of today's World Bank groups is a valid test of a contaminated measure. No standard error repairs that.

The question you did not ask. The claim is about whether conflict is *concentrated* in poorer countries. Our gap is one way of measuring that. Others would count countries, or deaths, or new conflicts (onsets) rather than ongoing ones. Each is a different question with its own test.

Key concept: inference handles chance; design handles everything else

Standard errors, intervals and p-values measure one thing: the luck of the draw. They say nothing about whether the comparison is fair, whether the measure captures the concept, or whether the arrow runs the way the title says. A perfectly executed test can answer the wrong question.

10 Reporting a test, and prompting for one

When you report a test, or check one your assistant ran, go through this list.

1. **The null hypothesis**, in words. ("Income has nothing to do with conflict.")
2. **The estimate**, in units a reader understands. ("28.5 percentage points.")
3. **The interval**, and how the standard error was computed. ("95% interval 18 to 39 points; standard errors clustered by country.")
4. **The p-value**, as a footnote to the interval. ("$p < 0.001$.")
5. **The unit and the number of independent pieces of evidence**. ("5,779 country-years from 167 countries.")

6. **The choices**, and whether the answer depends on them.

An assistant will run any test you ask for, and it will usually treat every row as independent unless told otherwise. A good prompt names the test, the unit, and the dependence:

```
Using data/lab2_full_table.csv, test whether the share of country-years in
conflict differs between the poorest and the richest quarter (income in the
previous year, quarters within each year, 1991-2025).  The same countries
appear in many years, so do not treat country-years as independent:  report
the gap, a 95% interval and a p-value from a permutation test that shuffles
whole countries, and also from standard errors clustered by country.  Print
every number; do not write any interpretation.
```

Then check the numbers it prints against the ones in this guide.

The R commands in this guide

Command	What it does
`sample(x)`	the values of `x` in random order (one shuffle)
`set.seed(13)`	makes the random shuffles repeatable
`mean(abs(shuffled) >= abs(obs))`	permutation p-value: share of shuffles at least as extreme
`prop.test(c(k1, k2), c(n1, n2))`	test and interval for the difference between two shares
`chisq.test(table(g, y))`	are the shares the same in all groups?
`t.test(x1, x2)`	test and interval for the difference between two means (Welch)
`lm(y ~ g)`	regression; with a 0/1 `g`, the slope is the difference in means
`coeftest(m, vcov = vcovCL(m, cluster = ~ id))`	the same with standard errors clustered by `id`
`qnorm(0.975)`	1.96: how many standard errors for a 95% interval
`2 * pnorm(-abs(z))`	two-sided p-value for a test statistic z

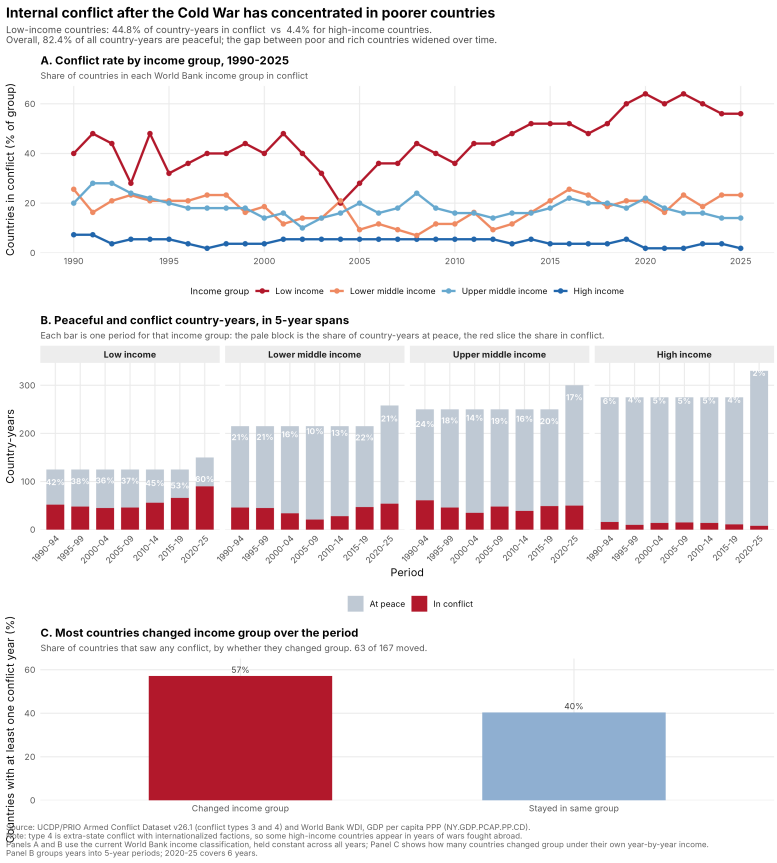
Decoder

Symbol or term	Meaning
H_0	the null hypothesis: "nothing going on"
H_1, H_A	the alternative hypothesis: "something going on"
$\hat{p}$	a share computed from the data (an estimate)
SE	standard error: how much an estimate moves from one run of history to the next
z, t	estimate divided by its standard error
p	how often a result at least this extreme would occur if H_0 were true
α	the false-alarm rate we accept, usually 0.05
χ^2, df	chi-square statistic; degrees of freedom
CI	confidence interval; "95%" describes the method, not this one interval
permutation test	a test that builds the null distribution by shuffling
null distribution	the results chance alone would produce
cluster	a group of related observations (here: all years of one country)
clustered (cluster-robust) SE	a standard error that allows observations within a cluster to be related
design effect	how much dependence inflates the variance compared with independent data
effective sample size	the number of independent observations the data are worth
Type I / Type II error	false alarm / missed effect
power	the chance of detecting an effect that is there
forking paths	choices made after seeing the data, which invalidate the p-value

Appendix: your figures

These are the figures that came back after Session 3, without names. I re-ran each script on `lab2_full_table.csv`, changing only file paths, so the numbers can differ slightly from the version you made on your own table. The notes say what each figure shows, what it does well, and what to look at again in the light of this guide. Each author has a fully annotated copy of their script.

Figure A

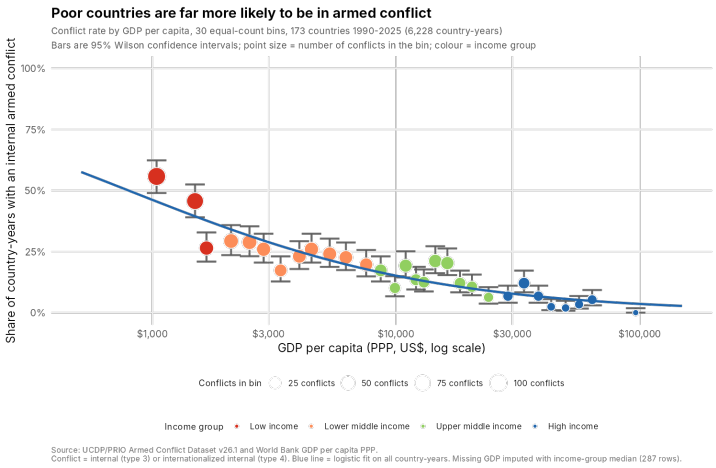


What it shows. Three panels: the share of country-years in conflict in each World Bank income group, year by year (A); peaceful and conflict country-years in five-year spans (B); and whether countries that changed income group had more conflict (C).

What works. Panel A compares shares within groups, with the peaceful years in the denominator: exactly the comparison of Section 1.1. The script says openly that the groups are today's classification.

Look again at. The headline "the gap widened over time": Section 7 tests it, and with today's groups the widening is driven by countries that became poor through recent wars. Panel B plots counts of country-years, not shares, and its percentage labels sit in the wrong block. Panel C applies today's World Bank thresholds for GNI (Atlas method) to GDP at PPP and to 1990 incomes. UCDP type 4 is internationalised internal conflict, not extra-state conflict.

Figure B1

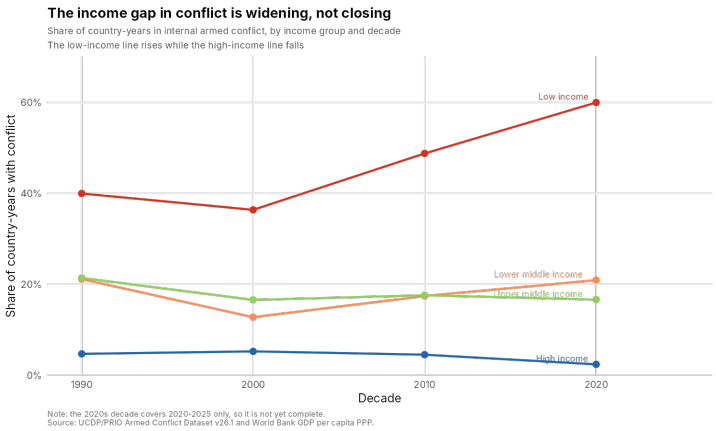


What it shows. The share of country-years in conflict in 30 equal-sized income bins, on a log income axis, with 95% intervals and a logistic curve.

What works. A log axis and binning are good answers to the question of how to plot a 0/1 outcome against income. It shows uncertainty, which none of the other figures do.

Look again at. The intervals treat 6,228 country-years as independent; Section 6 shows the honest ones are about four times wider. The 287 country-years without income were given the median income of their (current) income group, so some points rest on invented values. Thirty bins are thirty estimates: a few will stray by chance (Section 8).

Figure B2

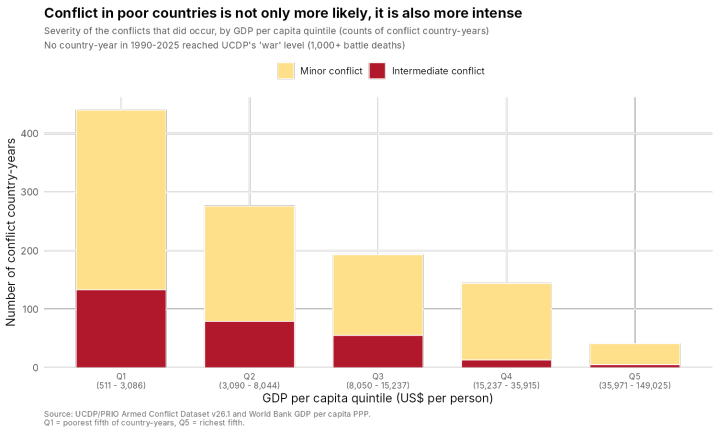


What it shows. The share of country-years in conflict by World Bank income group and decade.

What works. Shares, not counts; direct labels instead of a legend; a note that the 2020s are incomplete.

Look again at. The title "the income gap is widening". With today's income groups, Syria, Yemen, South Sudan, Mali and others count as low-income in every decade because their wars after 2010 made them poor. With income measured in the previous year, the gap changes by 4 points, well within chance (Section 7).

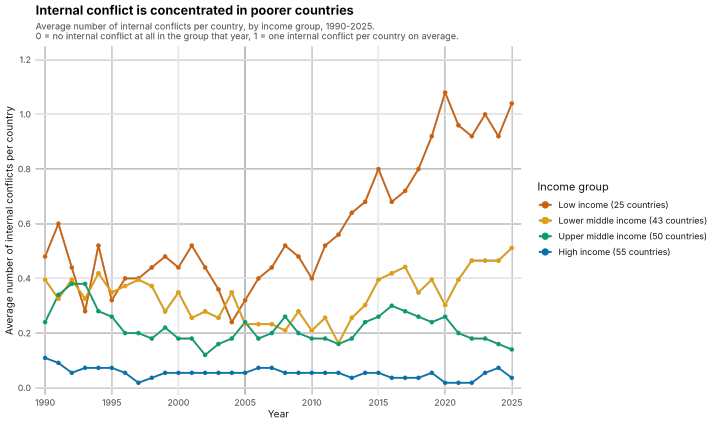
Figure B3



What it shows. Conflict country-years in five income bands, split by intensity.

Look again at. UCDP's intensity has two levels: 1 is minor (25–999 battle deaths in the year) and 2 is war (1,000 or more). So "Intermediate conflict" is war, and the subtitle ("no country-year reached war level") is false: there are 285 war country-years. The bars are counts, so the poorest band is tallest mainly because it has the most conflict years. For "more intense", compare the *share* of conflict years that are wars: about 30% in the three poorer bands, 9–12% in the two richest. Check codebook facts in the codebook.

Figure C

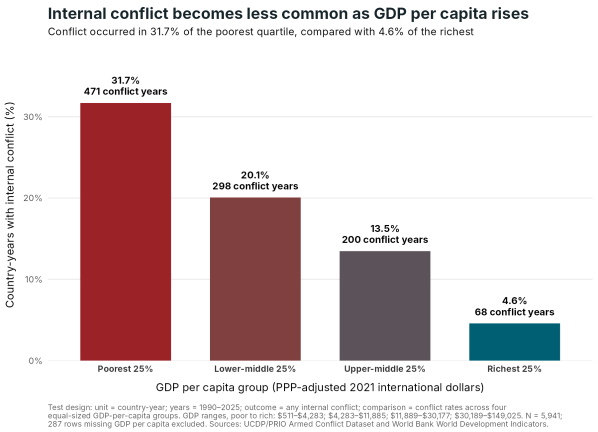


What it shows. The average number of internal conflicts per country, by World Bank income group, year by year.

What works. The most careful script: it stops if values are missing, if a country changes group, or if group sizes change, and it prints the years that break the pattern. The legend gives the number of countries behind each line.

Look again at. The measure counts conflicts, not countries in conflict: a country with four conflicts in a year counts four times (27% of conflict country-years have more than one). The claim is about where conflict happens, so the share in conflict is closer to it. "The poorest group is on top in every year" is a 36-test rule; Section 8 shows why a few exceptions say little.

Figure D

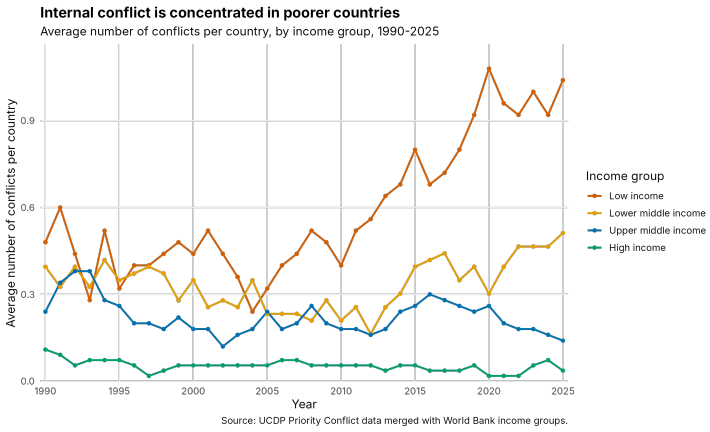


What it shows. The share of country-years in conflict in four equal-sized income groups.

What works. The clearest test of the five: one comparison, shares with the peaceful years included, missing income dropped and reported, the test design written into the caption.

Look again at. The four groups are cut over all years at once; incomes rose, so the poorest group is mostly early years. Cutting within each year compares countries with their contemporaries. Income is from the same year as the conflict. Here neither choice changes much (Figure 10): 27 points rather than 28.5.

Figure E



What it shows. The average number of conflicts per country, by World Bank income group, year by year: the same design as Figure C.

What works. Short, readable code that checks for missing values and a balanced panel.

Look again at. As for Figure C, the measure counts conflicts rather than countries in conflict, and the groups are today's. The colours break the poor-to-rich order (upper-middle is blue, high income green). The source is the UCDP/PRIO Armed Conflict Dataset, version 26.1; there is no "UCDP Priority Conflict" dataset.

References

Collier, Paul, and Anke Hoeffler. 2004. "Greed and Grievance in Civil War." *Oxford Economic Papers* 56 (4): 563–95.

Dean, Susan, and Barbara Illowsky. 2018. *Introductory Statistics*. Houston: OpenStax. Chapters 9–10.

Fearon, James D., and David D. Laitin. 2003. "Ethnicity, Insurgency, and Civil War." *American Political Science Review* 97 (1): 75–90.

Gelman, Andrew, and Eric Loken. 2014. "The Statistical Crisis in Science." *American Scientist* 102 (6): 460–65.

Gleditsch, Nils Petter, Peter Wallensteen, Mikael Eriksson, Margareta Sollenberg, and Håvard Strand. 2002. "Armed Conflict 1946–2001: A New Dataset." *Journal of Peace Research* 39 (5): 615–37.

Imai, Kosuke. 2017. *Quantitative Social Science: An Introduction*. Princeton: Princeton University Press. Chapter 7.

Kennedy, Ryan, and Philip D. Waggoner. 2021. *Introduction to R for Social Scientists: A Tidy Programming Approach*. Boca Raton: CRC Press. Chapter 7, sections 7.1–7.3.

Miguel, Edward, Shanker Satyanath, and Ernest Sergenti. 2004. "Economic Shocks and Civil Conflict: An Instrumental Variables Approach." *Journal of Political Economy* 112 (4): 725–53.

Wasserstein, Ronald L., and Nicole A. Lazar. 2016. "The ASA Statement on p-Values: Context, Process, and Purpose." *The American Statistician* 70 (2): 129–33.